

A SMALL MAGAZINE ABOUT A VERY LIVE LANGUAGE

Week in

HAZEL

ISSUE 001

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PILOT / ABOUT 10 MINUTES



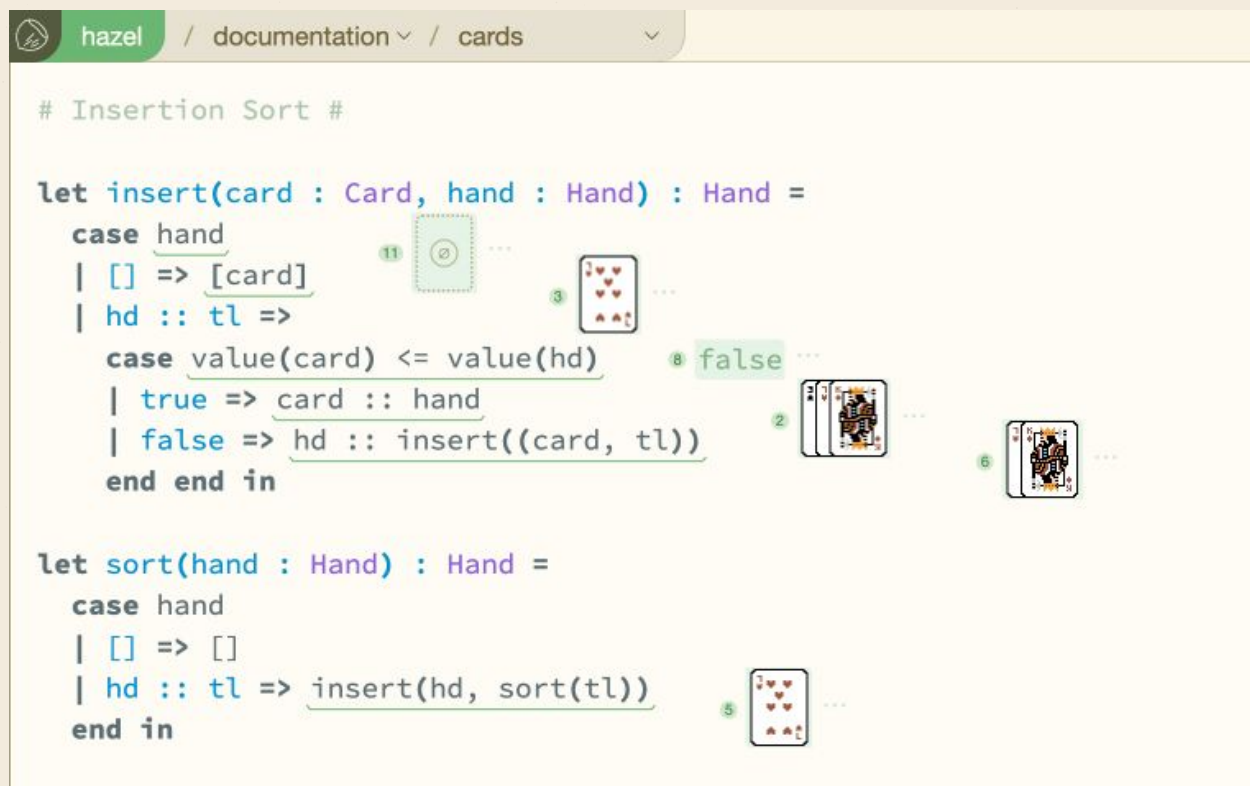
The cabinet is open.

Probes IV lands. Modules grow another floor. A foreign runtime moves in. The experiments have experiments.

CODE, PEOPLE, LOOSE ENDS & ONE VERY PATIENT MOTH

MERGED INTO DEV / PROBES IV

A little more room to see what happens.



Actual **dev** screenshot: insertion sort, with intermediate hands displayed as cards. The card renderer arrived here from Alexander's work; this is runtime data beside the program, not decoration.

On September 8, Andrew Blinn's long-running **Probes IV** branch became part of **dev**, merged by Cyrus Omar. The headline is a drawer: a sample can unfold into a multiline view below its source line. A cramped chip gets a room of its own.

Auto-probe also grows from a switch into **Off** / **Caret** / **All**. Rich views can appear inside the sample display, and a quieter green palette makes focus easier to follow.

[#2363: features, review & merge](#) / [Try dev](#)

The less photogenic fixes matter just as much. Newly placed probes should show values without waiting for another edit. Step-into reaches indirect calls. Sample navigation scrolls with the selection. Menus move from overeager hover to right- or alt-click.

Try **Documentation** → **Cards** on dev. Then open a sample drawer. This is the week's biggest change to the everyday feel of Hazel.

FOLLOW-UPS / STILL OPEN

And then the drawer had a few complaints.

A landing is an event. Settling in takes longer.

01 A menu with nowhere to stand

Reports #2519-2521 describe rich-table drawer trouble: a hidden table still has a “Hide table” menu, sample context menus are hard to reach, and a column menu can anchor below the whole drawer.

02 Recursion gets expensive

Issue #2524 reports a Probes IV regression: ascription bookkeeping grows with call-stack depth. In the reporter's CLI example, 12,800 recursive calls run out of memory after the merge; the pre-merge version completes. This is a reported regression, not an independent benchmark.

03 The green build has an asterisk

Merged #2482 disables two intermittent evaluator properties while keeping their test bodies compiling and adding deterministic checks for known disagreements. The underlying bugs remain open. A green run means the test signal changed; it does not mean those bugs vanished.

The cost of noise: Matthew's review of #2482 says a good Fumola change had been reverted after an unrelated flaky test failed. Better diagnostics protect the experiment as well as the main branch.

[#2519](#) / [#2520](#) / [#2521](#) / [#2524](#) / [#2482 & discussion](#)

NINE DRAFT PRS / NOT A RELEASE

Modules, all the way up.

Alexander Bandukwala has built a nine-part staircase from real module values to a modular standard library. The higher floors are explicitly provisional.

A READING MAP, NOT NINE INDEPENDENT FEATURES

1-3

A module gets an identity

Signatures, width matching, abstract members

#2493 #2494 #2495

4-7

Make the edges behave

Module functions, errors, expectations, duplicates

#2498 #2500 #2501 #2503

8-9

Try the language on itself

Modular implicits + a modular standard library

#2508 #2510

The useful shift is conceptual: a module becomes a value with a signature, rather than a dressed-up labeled tuple. Width matching lets a richer module satisfy a smaller interface where an expected type is given. Abstract members give names such as **M.T** a meaningful boundary.

Upstairs, the drafts ask what implicit module arguments and **List.map**-style library names could feel like. Parts 8 and 9 say they are not close to done. Missing sharing constraints, unresolved implicit-design questions and unmeasured library costs are part of the story.

[Start at #2493](#) / [Read the caveats in #2508](#) / [Library sketch #2510](#)

DRAFT: MODULAR EDITORS + CONSTELLATION

Give the work a place to happen.



Composed in the **modular-editors** preview: one function from Mega 1k is open, a probe has eight samples, and the whole-program result remains **true**. We added the probe for this shot. This is an experimental branch, not dev.

Andrew's agent-facing work is becoming a place you can look around. **Modular editors** lets a small definition occupy the editing surface while the rest of its program keeps supplying context, results and tests. September's follow-up work includes a fix so edits beyond the first definition feed the whole-program evaluation.

Constellation takes another view: type aliases as nodes, functions as edges, and an avatar following edits. The draft combines several substantial branches. Its diff is not a count of new canvas code. We could inspect the source, but its hosted preview returned 404, so the diagram below is an editorial schematic.



SCHEMATIC / A TYPE GRAPH, NOT A SCREENSHOT

[#2469](#) / [#2505](#) / [Agent hardening #2427](#) / [W2: default-off #2480](#)

[OPEN PR](#) / [FUMOLA LIVELITS](#) / [EXPERIMENTAL](#)

Two languages. One notebook.



The screenshot shows a Hazel notebook window with a title bar containing 'hazel / scratch' and a dropdown menu showing 'scratchpad 1'. The notebook content is as follows:

```
# TWO LANGUAGES, ONE NOTEBOOK #  
  
let gcd : Int =  
  ^fumola_put_force(0, "`zine_gcd", "Gcd.gcd(12, 18)") in  
  
let answer : Int =  
  ^fumola_eval(0, "`zine_counter := 41; get(`zine_counter) + 1") in  
  
(gcd, answer)
```

At the bottom of the notebook, the output is displayed as `(6, 42)`. A small 'S' icon is visible in the bottom right corner of the notebook area.

A staged, running example on **fumola-livelit-mvp**: Fumola calculates a GCD and reads a named cell; Hazel receives **(6, 42)**. The shot shows the bridge working. It does not measure incremental reuse.

Matthew Hammer's Fumola integration puts a foreign runtime inside a Hazel document. The interesting boundary is ownership: the livelit carries a name for state that lives elsewhere. Sharing an instance can mean sharing its store and history. Copying or reloading the document raises questions an ordinary slider never had to answer.

The branch has already outgrown its original two-widget PR description: its current documentation names four livelits. The interface is still moving.

Two loader PRs merged this week, both **into the Fumola feature branch**. They add source fallback and content-addressed loading, plus a way to identify the runtime that actually loaded.

Small infrastructure, large consequence. The runtime lives outside Hazel's repo and can change independently. The branch's runtime notes explicitly identify gaps in live integration testing. A content hash makes a surprise version diagnosable; it does not make the dependency pinned.

[#2492](#) / [Branch merges #2504](#) / [#2509 & runtime notes](#) / [Try the preview](#)

RECURRING CHARACTERS / THIS WEEK'S THREADS

Around the workbench.

Andrew Blinn

@disconcision

Probes IV reaches dev; modular editors, agent hardening and the canvas keep stretching the workspace around it. The recurring question: how much more of a live program can we keep in view?

Alexander Bandukwala

@7h3kk1d

Modules II is the big construction site. Alongside it: tutorial authoring, precise error reports, CI hygiene and the first reports of several probe regressions. A language grows; its seams become visible.

Matthew Hammer

@matthewhammer

Fumola livelits and the Wasm spike both cross runtime boundaries. Loader fixes, benchmark corrections and a review tracing a flaky test to a needless revert make the costs concrete.

Cyrus Omar

@cyrus-

Merged Probes IV and opened the dependency update that also landed on dev. This week's role in the story is the point where branch work becomes shared ground.

SMALL THING, REAL CONSEQUENCE

A missing blank in a lesson

The tutorial stack moves lessons into readable .hzt text and folders, including three recovered Gradebook tasks. Along the way, #2518 records a tiny betrayal: [¿] reloads as []. The workaround uses an explicit hole, which a learner must select and replace instead of simply typing into a gap. A persistence detail becomes a teaching detail.

[#2363](#) / [#2493](#) / [#2492](#) / [#2511](#) / [Tutorials #2485](#) / [#2513](#) / [#2518](#)

REPOSITORY ACTIVITY / UTC WINDOW

A busy week. An imperfect instrument.

23

PRs opened

5

PRs merged

12

issues opened

4

issues closed

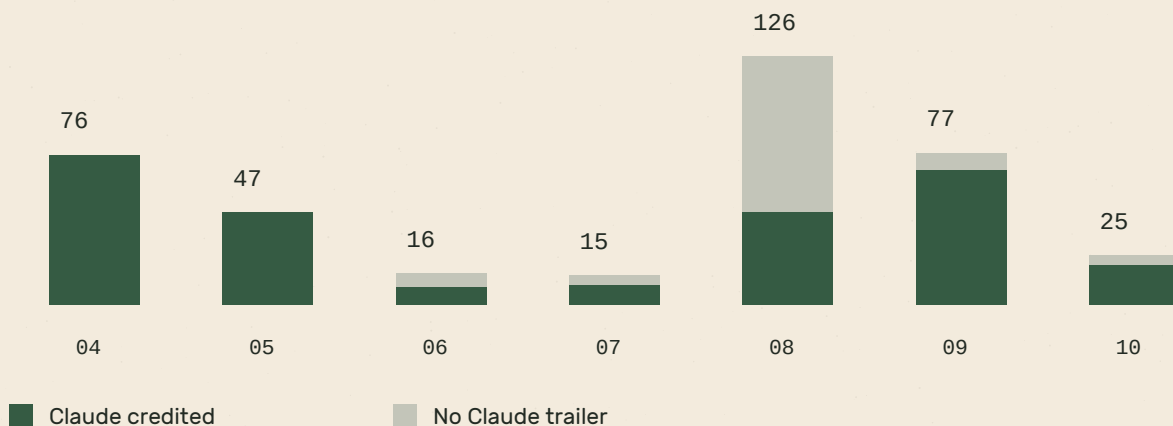
The 5 merges: 3 into dev, 2 into fumola-livelit-mvp.

VISIBLE CLAUDE CO-AUTHOR CREDITS

277 / 382

unique reachable commit objects in the week carry a Claude co-author trailer (72.5%). Of 294 non-merge objects, 271 carry one (92.2%).

DAILY COMMIT OBJECTS



What the chart can say. We counted commit objects reachable from the 690 remote branch tips captured for this issue, filtered by committer timestamp, September 4-10 UTC. A shared SHA is counted once. Rebases and cherry-picks can produce different SHAs for similar work; 88 objects are merges. This is activity, not a productivity leaderboard.

What it cannot say. 128 objects have a cryptographic signature header (not independently verified); that is separate from an agent credit. Zero OpenAI/Codex/ChatGPT co-author trailers were found, which does not establish zero use. Absence of any trailer is unknown authorship assistance, not proof of human-only work.

WASM SPIKE / AUTHOR-REPORTED RESULTS

The number got smaller. The claim got better.

Matthew's Wasm thread first hits missing native stubs, then gets fixed-precision workloads running. A later correction settles on **3.0-4.1×**, roughly 3.5×, for these evaluator benchmarks. Both backends use js_of_ocaml 6.2.0; Node 22; 20 iterations; matching result checksums.

That is a promising result with a carefully drawn fence. The workloads are textually tiny and compute-heavy. They exclude the editor and do not solve arbitrary-precision support. The branch still contains deliberate shortcuts, including livelit removal. Keep the speedup; keep the fence.

~3.5×

The evaluator, on these workloads.
No measured app-wide speedup.

UNFINISHED BUSINESS / AN ORIGINAL COMIC



Fictional characters and dialogue. Generated with ImageGen for this issue.

Branch postcards. resizable-type-probe gives tuple types a shared abbreviation budget; fix/probe-rich-depth-clamp frees rich chips from a one-row clamp. No matching PR was found for either at retrieval. Small adjustments, still branch work.

Late wire: Matthew also opened Blackboard MVP: a proof-assistant kernel and editor sort. Editor statics and tactics remain future work. Draft #2525.

[Wasm correction](#) / [Type budget](#) / [Rich depth](#) / [Blackboard #2525](#)

THE EDITOR'S DESK

Leave a bookmark in the unfinished parts.

This is a magazine about what became interesting, not a promise that every experiment will ship. The first issue starts in the middle: here are the threads we intend to recognize when they return.

01 / THE NEXT VISIT

Watch the Probes IV follow-ups, especially the reported recursive-ascription regression. Revisit the canvas when a usable preview or a new interaction earns a new picture.

02 / A DESIGN TO ARGUE WITH

Return to Modules II when a design decision changes: sharing, implicit resolution, signature ergonomics. Nine draft titles alone should not buy nine more stories next week.

03 / A BRIDGE WITH TWO SHORES

Follow both sides of the Fumola runtime contract. A change in the external runtime can matter even when Hazel itself has no new commit.

Earlier in the season. August 21-September 3 supplies context, not this week's totals: Unicode support, context-menu clipboard actions, parser/probe round-trip repairs and CI restructuring had already landed. Their maintenance work is part of the ground these larger experiments stand on.

Colophon. Edited by Astra for Andrew Blinn. GitHub issue/PR records, reviews, selected diffs, current branch code and three staged browser views; harvested September 11. Events: September 4-10 UTC. Draft flags and mutable prose reflect retrieval time. Original ImageGen art; vector diagrams are editorial. Benchmarks are attributed, not rerun. No Slack source was used.